

Multimedia Retrieval

Chapter 2: Performance Evaluation

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Which system is better?

Designing a benchmark

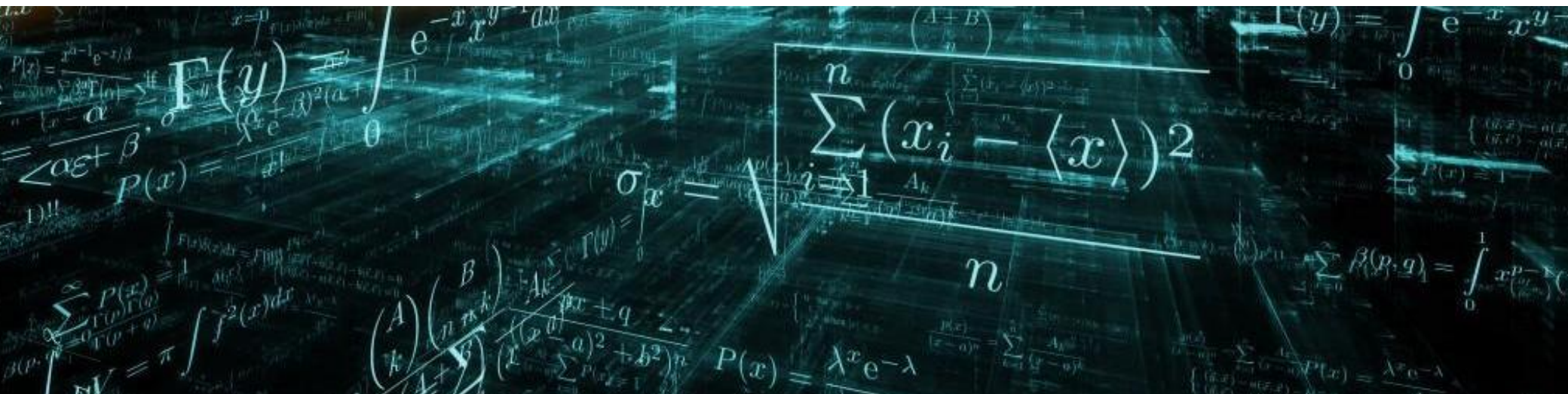
Precision and Recall

Ranked results

Graded relevance

Evaluating classifiers

Scores, thresholds, and ROC



Which system is better?

- Scenario:
 - Library with **50 books**
 - Student searches for "computer science foundation" literature
 - There are **15 relevant books** in the library
 - System A: returns **25 books** and finds **12 out of the 15 relevant books**
 - System B: returns **8 books** and finds **6 out of the 15 relevant books**
- Which system is good?
 - A
 - B
 - A and B
 - neither A nor B

System A	Relevant
Computer Organization & Design	Yes
Operating System Concepts	Yes
The Handmaid's Tale	No
Narrative of F. Douglass	No
Towards a New Architecture	No

System B	Relevant
Database Systems	Yes
Pattern Recognition & ML	Yes
Introduction to Algorithms	Yes
On the Origin of Species	No
Theory of Computation	Yes

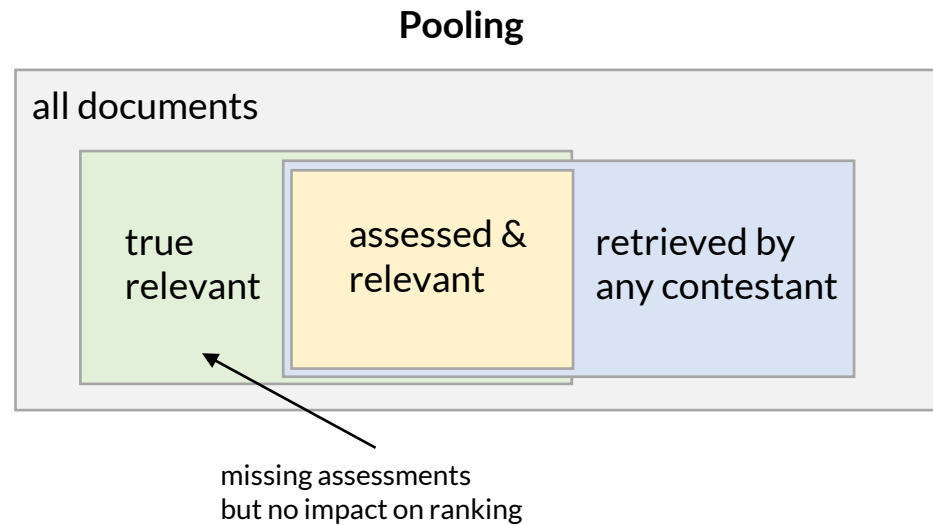
"Better" depends on the need

- Definition of "good" varies:
 - Fact check: read 2-3, stop → short clean list wins
 - Patent lawyer: miss nothing → broad coverage wins
 - Student: broad enough, short enough → neither serves well
 - Web search: top answers must be (highly) relevant
- But how do we measure this
 - Anecdotes are not evidence
 - One query where A beats B proves nothing
 - Need a fixed setup: collection, needs, judgments, procedure
 - Others rerun it → same numbers
- Relevance is a judgment about a need, not a property of a book (also not: contains the query keywords)
 - Information need != query, same query but different information need can lead to different results
 - Example: "Ford" (president or car?)



Designing a benchmark

- We need a test setup (benchmark) to measure:
 - **Collection:** the searchable collection, frozen, representative for real world scenario
 - **Needs:** what users want; separate from the query string
 - **Judgments:** the ground truth; written guidelines what is relevant, then frozen
 - **Procedure:** the rule turning results into a number; fixed in advance
- Key challenge: Judging at scale
 - Cannot judge 50M documents exhaustively
 - Judge only what some system retrieved; rest = not relevant
 - Ordering between systems survives; absolute numbers approximate
 - Bias: penalizes a novel system that finds unjudged relevant docs

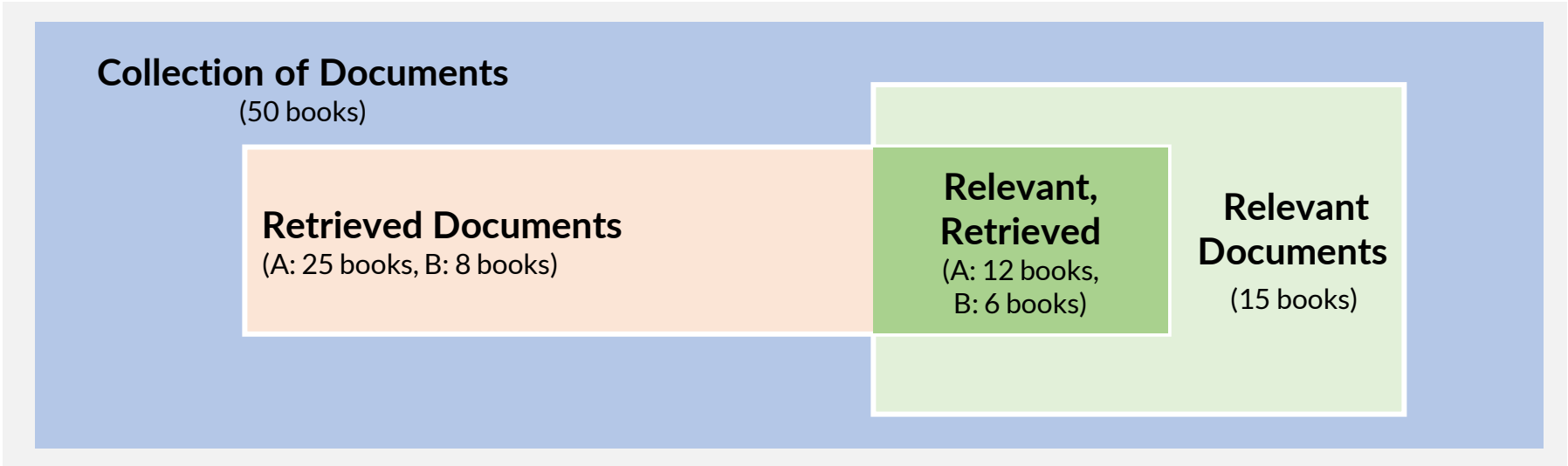


Counting Outcomes

- We first consider boolean results (no order)
- Every document → one of four outcomes
 - Dim 1: Retrieved vs. Not retrieved
 - Dim 2: Relevant vs Not relevant
- All set metrics derive from TP, FP, FN, TN
 - Retrieved = TP + FP
 - Relevant = TP + FN

	Relevant	Not relevant
Retrieved	True Positive (TP)	False Positive (FP)
Not retrieved	False Negative (FN)	True Negative (TN)

	TP	FP	FN	TN
System A	12	13	3	22
System B	6	2	9	33



Precision and Recall

- **Precision (p):** Of all retrieved results, how many are relevant?
- **Recall (r):** Of all relevant results, how many were retrieved?

- **Example: Precision and Recall for A and B**

- System A retrieved 25 books and found 12 of the 15 relevant ones
- System B retrieved 8 books and found 6 of the 15 relevant ones

- **Further definitions:**

- **Fallout (f):** how much of the irrelevant material got pulled in
- **F_β -measure:** weighted harmonic mean of precision and recall

$$p = \frac{TP}{TP + FP}$$

$$r = \frac{TP}{TP + FN}$$

$$p_A = \frac{12}{12 + 13} = \frac{12}{25} = 48\% \quad r_A = \frac{12}{12 + 3} = \frac{12}{15} = 80\%$$

$$p_B = \frac{6}{6 + 2} = \frac{6}{8} = 75\% \quad r_B = \frac{6}{6 + 9} = \frac{6}{15} = 40\%$$

$$f = \frac{FP}{FP + TN}$$

$$F_\beta = \frac{(\beta^2 + 1) \cdot p \cdot r}{\beta^2 \cdot p + r}$$

β	Weighting	$F_\beta(A)$	$F_\beta(B)$	Winner
0.5	favours precision	0.522	0.638	B
1.0	equal weight	0.600	0.522	A
2.0	favours recall	0.706	0.441	A

Averaging across many needs

- Benchmarks have multiple queries
 - **macro-averaging:** average precision / recall value over all queries
 - this is the standard for retrieval
- Example with 3 queries
- Less common: micro averaging (later this becomes more important)
 - more robust if some queries have few relevant document

$$p_{macro} = \frac{1}{N} \sum_{i=1}^N p_i$$

$$r_{macro} = \frac{1}{N} \sum_{i=1}^N r_i$$

Need	Relevant	System A: retrieved / found	System B: retrieved / found
CS foundations	15	25 / 12	8 / 6
Stage plays	5	6 / 4	4 / 3
Russian novels	1	3 / 0	1 / 1

$$r_{macro,A} = \frac{0.800 + 0.800 + 0.000}{3} = 53.3\%$$

$$r_{macro,B} = \frac{0.400 + 0.600 + 1.000}{3} = 66.7\%$$

$$p_{micro} = \frac{\sum_{i=1}^N T P_i}{\sum_{i=1}^N (T P_i + F P_i)}$$

$$r_{micro} = \frac{\sum_{i=1}^N T P_i}{\sum_{i=1}^N (T P_i + F N_i)}$$

What about the order?

- So far, we only looked at result sets
- But: in many use cases order of results matter
 - **web search**: want a relevant document at the top
 - **e-commerce**: most relevant products should be at the top (e.g., top 10 items)
- Look at the same example as before
 - how can we "encode" position into our metrics
 - what means "good" now?

#	System A	Relevant
1	Computer Organization & Design	Yes
2	Operating System Concepts	Yes
3	The Handmaid's Tale	No
4	Narrative of F. Douglass	No
5	Towards a New Architecture	No

#	System B	Relevant
1	Database Systems	Yes
2	Pattern Recognition & ML	Yes
3	Introduction to Algorithms	Yes
4	On the Origin of Species	No
5	Theory of Computation	Yes

Precision at Rank k

- The fraction of relevant documents among the first k results
 - only interested in top results (e.g., first page results)
 - relevant items after position k not important (and with that recall)

- Example

$$P@k = \frac{|\{D_1, \dots, D_k\} \cap Rel|}{k}$$

k	$P@k(A)$	$P@k(B)$
3	$2/3 = 66.7\%$	$3/3 = 100\%$
5	$2/5 = 40.0\%$	$4/5 = 80.0\%$
10	$5/10 = 50.0\%$	$6/10 = 60.0\%^*$

Mean Reciprocal Rank

- Rank of first relevant document, focuses only on top result
 - High importance on top position, quickly decreases for later ranks
 - Ignores everything after the first hit
- MRR = average of reciprocal ranks across all queries
- Example

$$RR = \frac{1}{\text{rank of first relevant document}}$$

$$MRR(Q) = \frac{1}{|Q|} \sum_{i=1}^{|Q|} RR_i$$

Query	System A: rank	RR_A	System B: rank	RR_B
“Who wrote Dracula?”	1	1.000	3	0.333
“A book on operating system design?”	4	0.250	1	1.000
“A nineteenth-century Russian novel?”	6	0.167	2	0.500
“A dystopian novel by Atwood?”	2	0.500	5	0.200

$$MRR_A = \frac{1.000 + 0.250 + 0.167 + 0.500}{4} = 0.479$$

$$MRR_B = \frac{0.333 + 1.000 + 0.500 + 0.200}{4} = 0.508$$

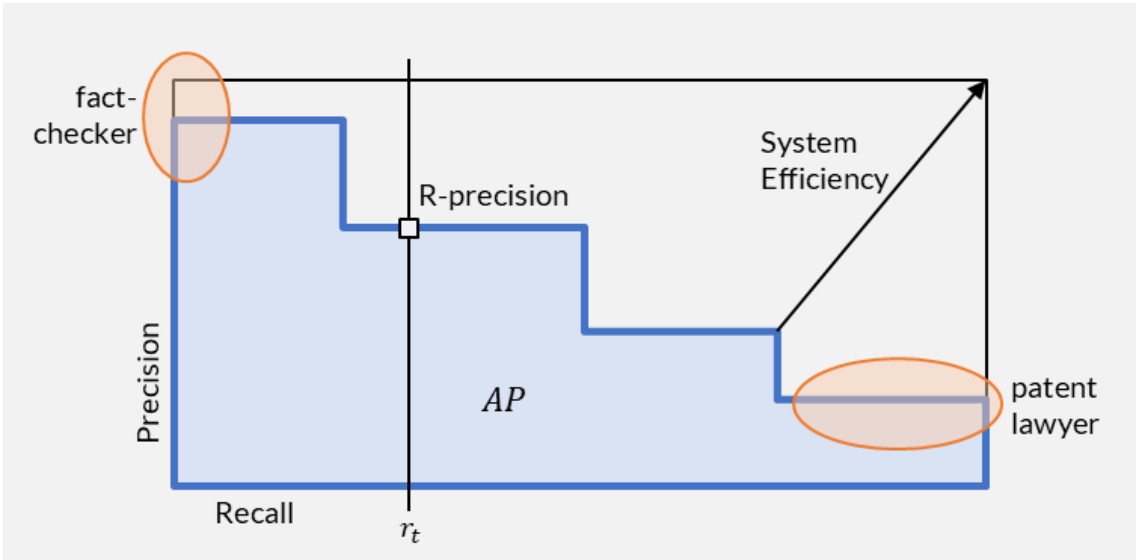
The Precision-Recall Curve

- One point at each relevant rank
 - Recall rises, precision usually falls
 - Upper-left = fact-checker
 - Lower-right = patent lawyer

- R-Precision
 - Precision after exactly ***R*** relevant documents have been retrieved

$$R-Prec = P@R = \frac{|\{D_1, \dots, D_R\} \cap Rel|}{R}$$

Rank	Relevant found	P_i	R_i
1	1	1/1 = 1.000	1/15 = 0.067
2	2	2/2 = 1.000	2/15 = 0.133
6	3	3/6 = 0.500	3/15 = 0.200
7	4	4/7 = 0.571	4/15 = 0.267
9	5	5/9 = 0.556	5/15 = 0.333
12	6	6/12 = 0.500	6/15 = 0.400
13	7	7/13 = 0.538	7/15 = 0.467
16	8	8/16 = 0.500	8/15 = 0.533
18	9	9/18 = 0.500	9/15 = 0.600
21	10	10/21 = 0.476	10/15 = 0.667
23	11	11/23 = 0.478	11/15 = 0.733
25	12	12/25 = 0.480	12/15 = 0.800



Mean Average Precision

- Area under the curve = Average Precision

$$AP = \frac{1}{|Rel|} \sum_{k: d_k \in Rel} P_k$$

- Example

$$AP_A = \frac{1}{15} \left(\frac{1}{1} + \frac{2}{2} + \frac{3}{6} + \frac{4}{7} + \frac{5}{9} + \frac{6}{12} + \frac{7}{13} + \frac{8}{16} + \frac{9}{18} + \frac{10}{21} + \frac{11}{23} + \frac{12}{25} \right) = 0.473$$

$$AP_B = \frac{1}{15} \left(\frac{1}{1} + \frac{2}{2} + \frac{3}{3} + \frac{4}{5} + \frac{5}{6} + \frac{6}{8} \right) = 0.359$$

- Mean Average Precision
 - Average of AP over all queries

$$MAP = \frac{1}{|\mathbb{Q}|} \sum_{i=1}^{|\mathbb{Q}|} AP_i$$

Graded relevance

- Not all relevance is equal
 - grades 0, 1, 2, 3 to measure **how** relevant
 - how does that change our metrics?
 - higher grades at higher ranks should count more than lower grades
 - AP: only relevance at high ranks preferred

- First idea:
 - Cumulative Gain (CG): sum up grades over top k ranks

Rank	System A	Grade
1	Computer Organization and Design	1
2	Operating System Concepts	2
3	The Handmaid's Tale	0
4	Narrative of the Life of Frederick Douglass	0
5	Towards a New Architecture	0

Rank	System B	Grade
1	Database Systems: The Complete Book	3
2	Pattern Recognition and Machine Learning	3
3	Introduction to Algorithms	1
4	On the Origin of Species	0
5	Introduction to the Theory of Computation	2

$$CG_k = \sum_{i=1}^k r\,el_i$$

$$CG_5(A) = 1 + 2 + 0 + 0 + 0 = 3$$

$$CG_5(B) = 3 + 3 + 1 + 0 + 2 = 9$$

Discounted Cumulative Gain

- Cumulative Gain does not take position into account, only how relevant documents are
- Discounting idea
 - Lower ranked document count less
 - Many ways how to discount, right side shows one common approach
- Issue
 - DCG values depend on the number of relevant documents that exist
 - Can vary greatly across different queries disallowing direct comparison
- Normalized DCG
 - Compute DCG for ideal ranking (=IDCG) and use this value to normalize DCG
 - nDCG: "how close to perfect ranking"

$$DCG_k = \sum_{i=1}^k \frac{rel_i}{\log_2(i+1)}$$

$$nDCG_k = \frac{DCG_k}{IDCG_k}$$

$$DCG_5(A) = 1.000 + 1.262 + 0 + 0 + 0 = 2.262$$

$$DCG_5(B) = 3.000 + 1.893 + 0.500 + 0 + 0.774 = 6.1665$$

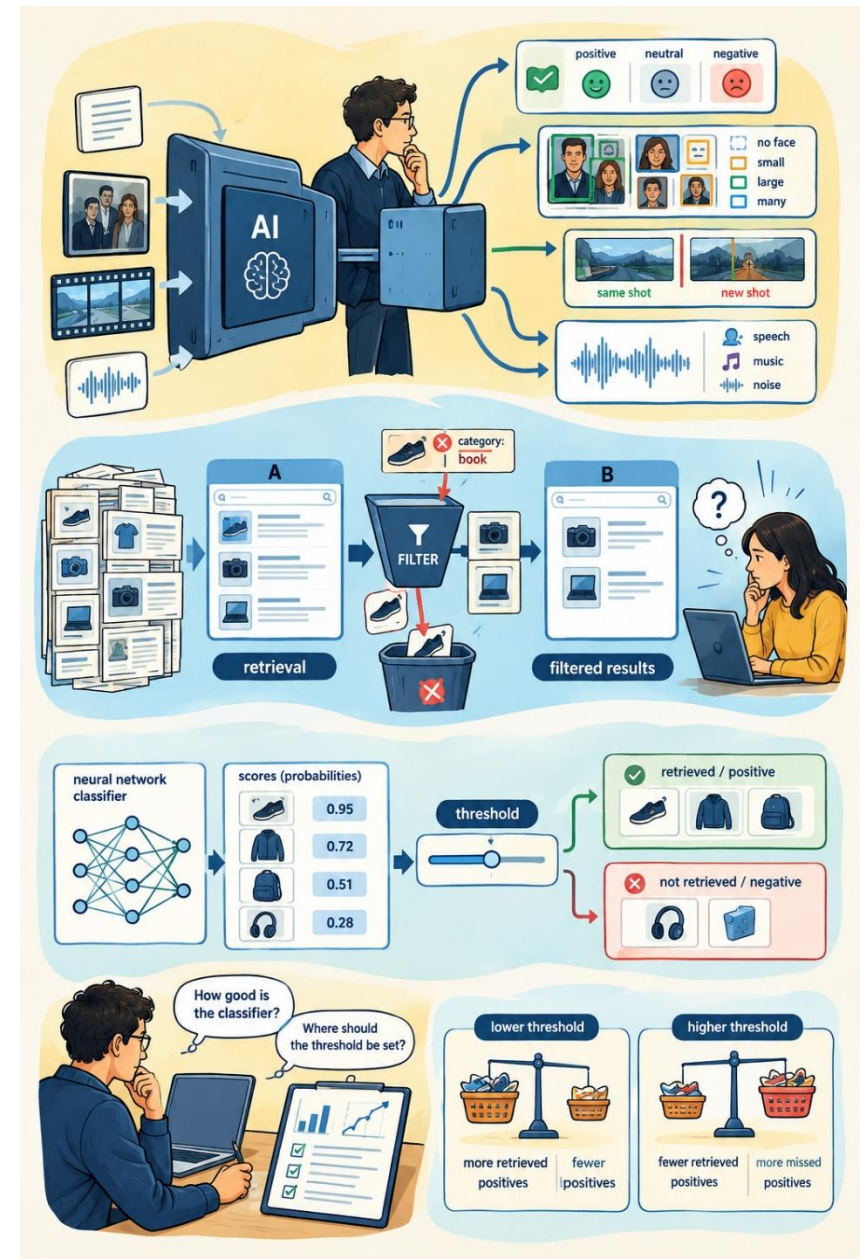
$$IDCG_5 = 8.028$$

$$nDCG_5(A) = \frac{2.262}{8.028} = 0.282$$

$$nDCG_5(B) = \frac{6.1665}{8.028} = 0.768$$

Evaluating Classifiers

- Retrieval pipelines rely on classifiers
 - Sentiment (positive / negative / neutral)
 - Face detection (none / small / large / many)
 - Shot segmentation (new shot?)
 - Audio (speech / music / noise)
- Filtering on metadata
 - Wrong label → relevant doc never shows
- Same question as precision/recall
 - Is the classifier good?
 - How to tune score-based classifiers (softmax $\neq$ 0/1)?



Confusion Matrix

- 2×2: rows = predicted, columns = ground truth
- Four counts: TP, FP, FN, TN
- Many metrics beyond precision/recall
- Accuracy: how often the label is correct
- Reading the matrix
 - Rate = cell ÷ row or column total ($PPV = TP / (TP + FP)$, $TPR = TP / (TP + FN)$)
 - Accuracy = $(TP + TN) / total$

		Actual Condition (as observed)			
		Population	Positive (P)	Negative (N)	
Predicted Condition (as computed)	True (T)	True Positive (TP)	False Positive (FP)	Positive Predictive Value (PPV), Precision	False Discovery Rate (FDR)
	False (F)	False Negative (FN)	True Negative (TN)	False Omission Rate (FOR)	Negative Predictive Value (NPV)
		True Positive Rate (TPR), Sensitivity, Recall, Hit Rate	False Positive Rate (FPR), Fall-Out	Accuracy (ACC)	
		False Negative Rate (FNR), Miss Rate	True Negative Rate (TNR), Specificity		Error Rate (ERR), Misclassification Rate

When Accuracy Misleads

- Rare disease: 30 sick in 2030 (1.5%)
- Classifier A: accuracy 90.6% – but precision 10% (9 of 10 positives healthy)
- "Always healthy": accuracy 98.5% – beats A, detects nothing
- Prevalence is the signal
 - Imbalanced classes (30 vs. 2000)
 - <10% or >90% prevalence → distrust accuracy
 - Pick metrics that rule-in / rule-out

Classifier A				
		Actual Condition (as observed)		
		Population (2030)	Positive ($P = 30$)	Negative ($N = 2000$)
Predicted Condition (as computed)	True (200)	True Positive ($TP = 20$)	False Positive ($FP = 180$)	$PPV = \frac{20}{200} = 10\%$
	False (1830)	False Negative ($FN = 10$)	True Negative ($TN = 1820$)	$NPV = \frac{1820}{1830} = 99\%$
		$TPR = \frac{20}{30} = 67\%$	$TNR = \frac{1820}{2000} = 91\%$	$ACC = \frac{1840}{2030} = 91\%$

Classifier B					
		Actual Condition (as observed)			
		Population (2030)	Positive ($P = 30$)	Negative ($N = 2000$)	
Predicted Condition (as computed)	True (0)	True Positive ($TP = 0$)	False Positive ($FP = 0$)	$PPV = \frac{0}{0} = n/a$	
	False (2030)	False Negative ($FN = 30$)	True Negative ($TN = 2000$)	$NPV = \frac{2000}{2030} = 99\%$	
		$TPR = \frac{0}{30} = 0\%$	$TNR = \frac{2000}{2000} = 100\%$	$ACC = \frac{2000}{2030} = 99\%$	

Multiclass Example: Intent Routing

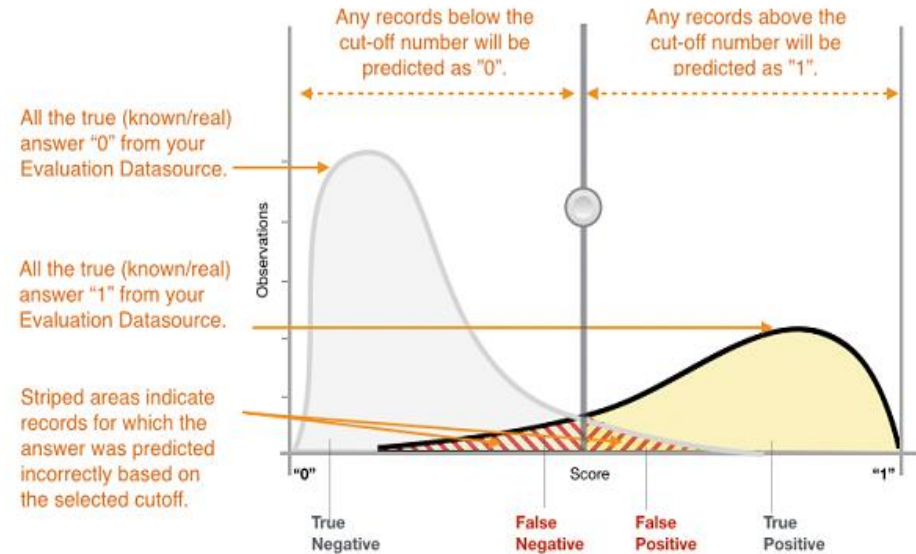
- Classify intent of user query
 - each intent follows different evaluation flow
 - frequently used in agentic systems
 - also common in web search
- Given 4 intents, how do we measure?
 - Use per-class metrics (one-vs-rest)
 - micro-average: e.g., accuracy is sum of correct over total, not macro-average over per-intent accuracy

		Actual Class			
Queries (200)		KB (80)	Web (60)	Calculator (40)	Clarify (20)
Predicted Class	KB (92)	70	8	2	12
	Web (60)	4	48	6	2
	Calculator (36)	2	3	30	1
	Clarify (12)	4	1	2	5

		Actual Class		
		Queries (200)	Clarify (20)	Not Clarify (180)
Predicted Class	Clarify (12)	5	7	precision PPV = 42%
	Not Clarify (188)	15	173	NPV = 92%
		TPR = 25%	TNR = 96%	ACC = 94%
		recall	specificity	accuracy

Score Distributions and Optimal Classification Threshold

- Binary decision from a continuous score
- Threshold to left → more "positive":
 - $\uparrow$ TP $\downarrow$ FN $\uparrow$ FP $\downarrow$ TN
- Threshold to right → more "negative":
 - $\downarrow$ TP $\uparrow$ FN $\downarrow$ FP $\uparrow$ TN
- FP vs FN is a fundamental trade-off
 - No threshold gives FP=FN=0
 - Choose: rule-in (low FPR) or rule-out (high TPR)

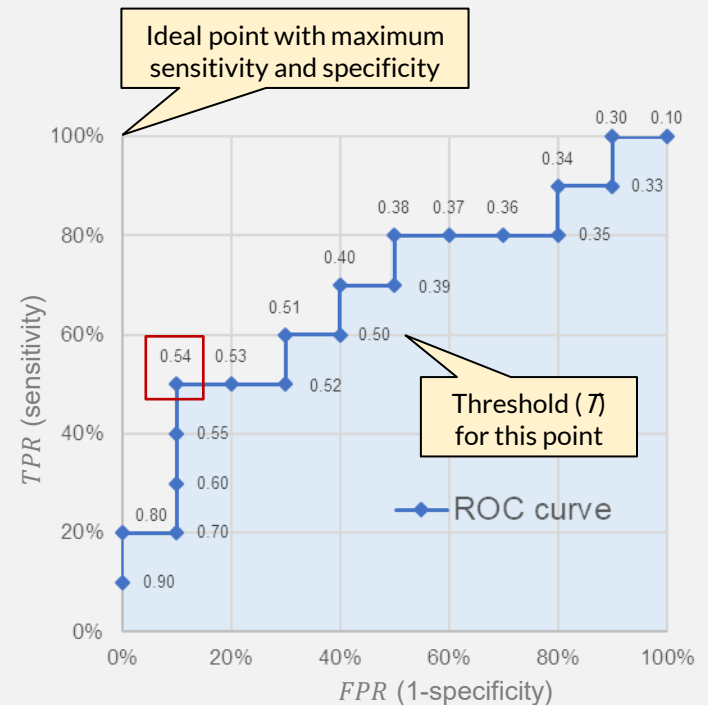


source: <https://docs.aws.amazon.com/machine-learning/latest/dg/binary-classification.html>

The ROC Curve

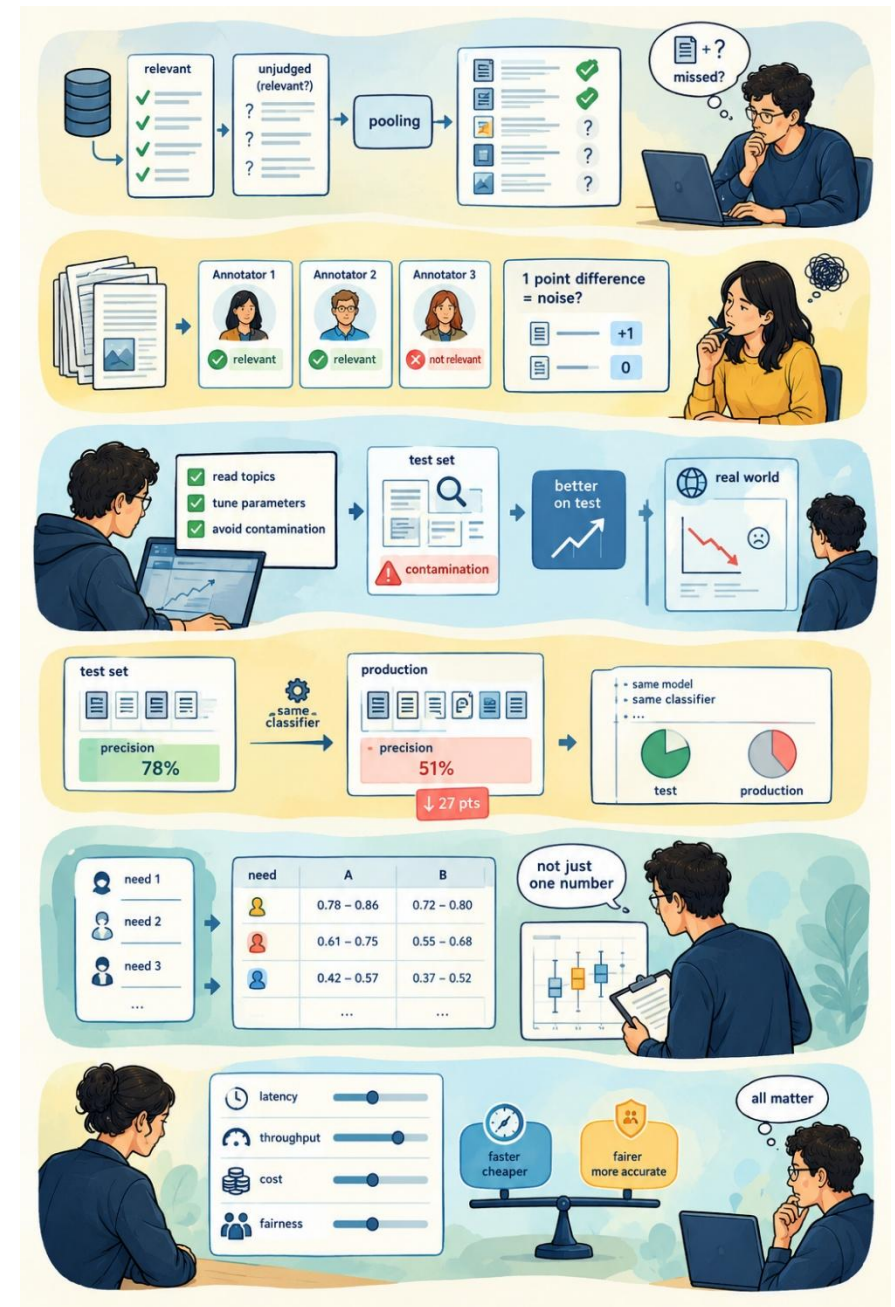
- Plot TPR and FPR as a function of the threshold
 - TPR = sensitivity = recall
 - FPR = 1-specificity
- Order test sample by score to obtain the graph
 - Choose threshold that optimizes objective (e.g., accuracy)
 - Area under curve (AUC) summarizes how well the classifier works across all threshold values

Class	Score	TP	FP	FN	TN	sensitivity	1-specificity	ACC
						TPR	FPR	
Spam	0.90	1	0	9	10	10%	0%	55%
Spam	0.80	2	0	8	10	20%	0%	60%
Legitimate	0.70	2	1	8	9	20%	10%	55%
Spam	0.60	3	1	7	9	30%	10%	60%
Spam	0.55	4	1	6	9	40%	10%	65%
Spam	0.54	5	1	5	9	50%	10%	70%
Legitimate	0.53	5	2	5	8	50%	20%	65%
Legitimate	0.52	5	3	5	7	50%	30%	60%
Spam	0.51	6	3	4	7	60%	30%	65%
Legitimate	0.50	6	4	4	6	60%	40%	60%
Spam	0.40	7	4	3	6	70%	40%	65%
Legitimate	0.39	7	5	3	5	70%	50%	60%
Spam	0.38	8	5	2	5	80%	50%	65%
Legitimate	0.37	8	6	2	4	80%	60%	60%
Legitimate	0.36	8	7	2	3	80%	70%	55%
Legitimate	0.35	8	8	2	2	80%	80%	50%
Spam	0.34	9	8	1	2	90%	80%	55%
Legitimate	0.33	9	9	1	1	90%	90%	50%
Spam	0.30	10	9	0	1	100%	90%	55%
Legitimate	0.10	10	10	0	0	100%	100%	50%



When evaluation misleads

- Incomplete ground truth:
 - pooling scores unjudged relevant docs as misses
- Noisy judgments:
 - annotators disagree — a one-point gap may be noise
- Optimizing against the test:
 - reading topics, tuning, contamination
- Test set ≠ production:
 - same classifier, shifted prevalence → precision drops 27pts
- One number is not a result
 - report the spread across many needs
- Effectiveness is one axis:
 - latency, throughput, cost, fairness also count



Further Reading

- Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to information retrieval*, Chapter 8: Evaluation in information retrieval. Cambridge University Press. [Read online](#). A concise foundation for precision, recall, MAP, and benchmark design.
- Järvelin, K., & Kekäläinen, J. (2002). *Cumulated gain-based evaluation of IR techniques*. *ACM Transactions on Information Systems*, 20(4), 422–446. [ACM record](#). The paper that introduced DCG and nDCG for graded relevance evaluation.
- Fawcett, T. (2006). *An introduction to ROC analysis*. *Pattern Recognition Letters*, 27(8), 861–874. [Read the paper](#). A practical guide to ROC curves, AUC, and threshold selection for binary classifiers.
- Armstrong, T. G., Moffat, A., Webber, W., & Zobel, J. (2009). *Improvements that don't add up: Ad-hoc retrieval results since 1998*. In *Proceedings of CIKM 2009*(pp. 601–610). [ACM record](#). An influential warning that static benchmarks can conceal a lack of cumulative progress.
- [TREC \(Text REtrieval Conference\)](#). NIST's continuing evaluation campaigns provide the pooled collections, shared topics, and comparative methodology behind much of this chapter's retrieval evaluation practice.